Research in Autism Spectrum Disorders 108 (2023) 102257



Contents lists available at ScienceDirect

Research in Autism Spectrum Disorders

journal homepage: www.elsevier.com/locate/rasd





Early childhood interventions in educational settings that promote school readiness for children with autism and other developmental disabilities: Systematic review

Zuyi Fang [a,*,1], Xinran Liu [a], Cheng Zhang [b], Dongping Qiao [a]

[a] School of Social Development and Public Policy, Beijing Normal University, No.19, Xinjiekouwai St, Haidian District, Beijing 100875, China
[b] School of Sociology and Social Policy, University of Leeds, Woodhouse Lane, Leeds LS2 9JT, UK

ARTICLE INFO

Keywords:
Autism
Developmental disability
School readiness
Education
Early Childhood Development
Meta-analysis

ABSTRACT

Background: Promoting school readiness is a vital strategy to close the learning gaps and reach children's full potential. This systematic review aims to examine the effectiveness of early childhood interventions delivered in educational settings for children with autism and other developmental disabilities aged 3–6 years in improving six developmental outcome domains.
Methods: Sensitive searches were conducted in seven international databases and three Chinese regional databases, as well as seven grey literature repositories, for all available studies published in English or Chinese language. All records were double screened. Two bilingual reviewers were involved in extraction and quality appraisal. Effect sizes were synthesised using robust variance estimation.
Results: Twenty-eight studies were included in the statistical analyses. Findings provided tentative evidence of social and communication trainings in improving social and communication skills (d = 1.22, 95% CI [0.68, 1.76]) and language interventions in increasing language abilities (d = 0.55, 95% CI [0.33, 0.76]). Treatment effects varied by diagnosis and context. There was substantial heterogeneity across studies.
Conclusion: We call for intersectional partnerships to promote the inclusion of children with autism and other developmental disabilities in early childhood service and education, as well as more robust research, especially in low- and middle-income countries and involving more types of developmental disabilities.

1. Developmental disabilities

Developmental disabilities are a set of severe chronic conditions that affect children's physical, cognitive, sensory, language, and behavioral functioning (Centers for Disease Control and Prevention, 2022). Global data on the number of children with developmental disabilities are lacking. According to Olusanya et al. (2020), an estimated 291.2 million children and adolescents worldwide are affected by epilepsy, intellectual disability, vision impairment, or hearing loss. Furthermore, 52.9 million children under 5 years of age have been diagnosed with autism, attention deficit hyperactivity disorder (ADHD), or the above four types of disabilities (Olusanya et al., 2018). These numbers could potentially be higher if we consider a broader range of developmental disabilities. Of these children,

* Correspondence to: School of Social Development and Public Policy, Beijing Normal University, Beijing, China.
E-mail address: fangzuyi@bnu.edu.cn (Z. Fang).
[1] https://orcid.org/0000-0002-8619-0137

https://doi.org/10.1016/j.rasd.2023.102257
Received 8 February 2023; Received in revised form 6 June 2023; Accepted 29 September 2023
Available online 10 October 2023
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95% reside in low- and middle-income countries (Olusanya et al., 2018) and the prevalence is likely to increase among impoverished families (Banks, Kuper, & Polack, 2017).

The impacts of autism and other developmental disabilities can extend beyond functional limitations. Children with developmental disabilities in resource limited settings can face exclusion from early childhood interventions and routine healthcare services (Walker et al., 2011), therefore receiving less stimulation and being ill-prepared for structured learning settings. Furthermore, the odds of experiencing all forms of violence are two times higher in children with disabilities than in non-disabled children, exacerbating the situation and further limiting their opportunities to learn (Fang, Cerna-Turoff, Zhang, Lu, Lachman & Barlow, 2022). This is primarily due to the insufficient service capacity, especially in low- and middle-income countries, and the lack of rigorous evidence guiding the adaptation and delivery of early childhood interventions for children with developmental disabilities (Grantham-McGregor, Cheung, Cueto, Glewwe, Richter & Strupp, 2007).

2. School readiness

School readiness emphasizes the importance of individual and environmental competencies to facilitate the transition from preschool to primary school settings (Bronfenbrenner, 1979). Six interrelated key domains, including cognitive development, social and communication skills, physical and motor development, speech and language skills, emotional and behavioral adjustment, and adaptive living skills, are essential for children to be ready for school (Snow, 2010; Williams et al., 2019). The minimum skills required may also vary across countries and cultures (Lynch & Soni, 2021).

Improving child school readiness has been identified as a crucial strategy for promoting educational and social equality. Research has shown that children who are prepared for school have better academic performance, higher school completion rates (Arnold, Bartlett, Gowani & Merali, 2007; Duncan et al., 2007; McCoy et al., 2017; Nonoyama-Tarumi & Bredenberg, 2009; Watts, Gandhi, Ibrahim, Masucci & Cybele Raver, 2018), and long-term benefits such as increased likelihood of attending college, being employed, and having fewer health problems as adults (Campbell, Ramey, Pungello, Sparling & Miller-Johnson, 2002; Heckman, Moon, Pinto, Savelyev & Yavitz, 2010; Reynolds, Temple, Ou, Arteaga & White, 2011; Reynolds et al., 2007). These improvements can promote social justice, accumulate human capital, and advance overall societal development (Sayre, Devercelli, Neuman & Wodon, 2015), particularly among vulnerable groups.

3. Preschool interventions

Preschool programs are crucial for promoting school readiness. Some examples of those programs are the Chicago School Readiness Project, the Child-Parent Center Education Program, the Abecedarian Project, and the HighScope Perry Preschool Program in the United States (Campbell et al., 2002; Heckman et al., 2010; Raver, Jones, Li-Grining, Zhai, Bub & Pressler, 2011; Raver, Jones, Li-Grining, Zhai, Metzger & Solomon, 2009; Reynolds et al., 2011; Watts et al., 2018; Webster-Stratton & Reid, 2004; Zhai, Raver, Jones, Li-Grining, Pressler & Gao, 2010), as well as the Incredible Years Training Series, the PLAYTOD program, and the Madrasa Early Childhood Development Program in countries of Europe and Africa (Baker-Henningham & Walker, 2009; Malmberg, Mwaura, & Sylva, 2011; Toussaint, Streppel, Mul, Fukkink, Weijs & Janssen, 2020; Webster-Stratton & Reid, 2004). Systematic reviews have shown that preschool interventions can effectively improve cognitive and socio-emotional outcomes for non-disabled children (Barnett, 2011; Camilli, Vargas, Ryan & Barnett, 2010) and increase their school retention and graduation rates (McCoy et al., 2017). There is also some evidence of substantial long-term positive effects (Barnett, 2011). However, fadeout patterns have also been observed in some interventions (Bailey, Duncan, Odgers & Yu, 2016).

There is also a growing number of preschool interventions implemented for children with autism and other developmental disabilities. For example, the Joint Attention Symbolic Play Engagement Regulation (JASPER) program has been shown to be effective in improving social-communication skills in children with autism (Chang, Shire, Shih, Gelfand & Kasari, 2016; Goods, Ishijima, Chang & Kasari, 2013; Lawton & Kasari, 2012). The Kids in Transition to School intervention has also demonstrated effectiveness in promoting cognitive skills and literacy among children with various developmental disabilities (McDermott et al., 2018; Pears, Kim, Fisher & Yoerger, 2016; Pears, Kim, Healey, Yoerger & Fisher, 2015). Additionally, incorporating Shared Book Reading into preschool curricula has been found to improve child language skills in children with learning disabilities (Crain-Thoreson & Dale, 1999; Dawes, Leitão, Claessen & Kane, 2019; Van Kleeck, Vander Woude, & Hammett, 2006).

Despite an increase in experimental studies focused on preschool children with autism and other developmental disabilities, few systematic reviews have been conducted to synthesize the evidence. The What Works Clearing House conducted a review on early childhood education interventions for preschool children with disabilities, but it was limited to children residing in the United States (What Works Clearinghouse, 2010b). There is a lack of reviews of such interventions delivered in other countries.

4. Objective

The purpose of this global review is to fill the aforementioned gap. It aims to understand the effectiveness of early childhood interventions delivered worldwide in educational settings for preschool children with autism and other developmental disabilities aged 3–6 years. The results of this review could be used to inform decision-making and strategic deployment of limited resources.

5. Methods

This review was prospectively registered in PROSPERO (CRD42022320519). We followed the Cochrane guidelines (Higgins & Green, 2011) for conducting this review and PRISMA guidelines (Page et al., 2021) for reporting the results.

5.1. Search strategy

On April 8th, 2022, we conducted sensitive searches in seven international databases (MEDLINE, Embase, PsycINFO, CINAHL, Child Development & Adolescent Studies, ERIC, and Cochrane Library) and three Chinese regional databases (China National Knowledge Infrastructure, China Science and Technology Journal Database, and Wanfang) for all available studies published in English or Chinese language. Sensitive searches are comprehensive and thorough searches that involve incorporating a wide range of relevant terms. The purpose of these searches is to capture as many necessary and relevant studies as possible in the review. Seven grey literature repositories and the reference lists of all included studies were also hand searched. Our search terms combined free-text and medical subject headings and were structured around the concepts of developmental disability, school readiness, and randomized controlled trial. We did not include terms to specify the type of intervention because the search strategies were also designed to identify studies that assessed the effectiveness of early childhood parenting interventions (Fang et al., 2023) and therapies provided in clinical settings (report in preparation).

5.2. Inclusion and exclusion criteria

We searched for early childhood interventions that aimed to promote school readiness for children with autism and other developmental disabilities. We imposed no restrictions on the types of developmental disabilities. Examples of developmental disabilities are autism, ADHD, intellectual disability, cerebral palsy, epilepsy, learning disability, language disorder, vision loss, and hearing loss. The diagnosis should be made by a professional through reliable screening and assessment. Children who were deemed at risk for developmental disabilities without a formal diagnosis were excluded. Studies that involved both children with and without developmental disabilities should provide separate analysis for children with developmental disabilities, unless children with developmental disabilities constituted more than 50% of the sample. Studies were eligible for inclusion if participants aged between 3 and 6. The age range of 3–6 years was determined by considering regional differences in enrolment in preschool education systems. Children who were older than 6 years were eligible for inclusion *only* if they were enrolled in preschool programs and had not yet entered higher-level education. When both preschool and older children were involved, preschool children should constitute at least 50% of the sample, or preschool children should be analyzed separately.

We included early childhood interventions delivered in educational settings, such as preschools, as part of the curriculum and excluded those that solely provided medication or medical treatment. The interventions could be cognitive, language, social-communication, or movement interventions, as well as comprehensive interventions that targeted multiple developmental domains. Studies must assess at least one of the six developmental domains associated with children's readiness for school, including child cognitive development; speech and language development; emotional and behavioral adjustment; social and communication skills; motor and physical development; and adaptive living skills. We included only randomized controlled trials (RCTs) and cluster-RCTs with an active (an alternative intervention, another school readiness intervention, or a variant of the school readiness intervention) or inactive control (service-as-usual, waitlist, or no treatment). Studies were excluded if more than half of the control group were children without developmental disabilities or having completely different diagnoses from those in the treatment group. No restrictions were imposed on geographical locations.

5.3. Study selection, data extraction, and quality assessment

Prior to screening, we removed duplications using EndNote X9 and Rayyan. ZF and CZ, who are bilingual in Chinese and English, independently screened all titles and abstracts and subsequently retrieved full texts to determine eligibility for inclusion. Disagreements were reconciled by discussion with XL. Cohen's Kappa coefficients were calculated to measure reviewer agreement.

Two bilingual authors independently extracted data using a standardized form, which included the following information: 1) basic study information (author, year, trial country/region, income level, world region); 2) program (program brand and description; staff qualification and training; session number, duration, and frequency); 3) study design (intraclass correlation coefficient and control condition); 4) participant (child age, gender, ethnicity, diagnosis, and family socioeconomic status); 5) outcome measure (outcome, measure, and reporting source); and 6) outcome data (sample size, number of dropouts, mean, and standard deviation in each treatment group at each time point).

Quality assessment was informed by the Cochrane Collaboration's tool for assessing risk of bias (Higgins & Green, 2011) and included quality domains related to random sequence generation, allocation concealment, blinding, incomplete outcome data, selective reporting, and other source of bias, particularly baseline group comparability and data dependency. We also assessed four additional items concerning intent-to-treat analysis, program developer involvement, trial protocol registration, and observational measures. Items were scored 1 or 0 for the presence or absence of a quality, except for developer involvement, which was scored 0–3 for developer being involved as researcher, program implementer, advisor, and not involved. The total score was 12. XL extracted and assessed the quality of 90% of included studies and ZF did the remaining 10%. Data extraction and quality assessment sheets were then exchanged to check entries for accuracy. Any disagreements were resolved in discussion. Cohen's Kappa coefficients were calculated

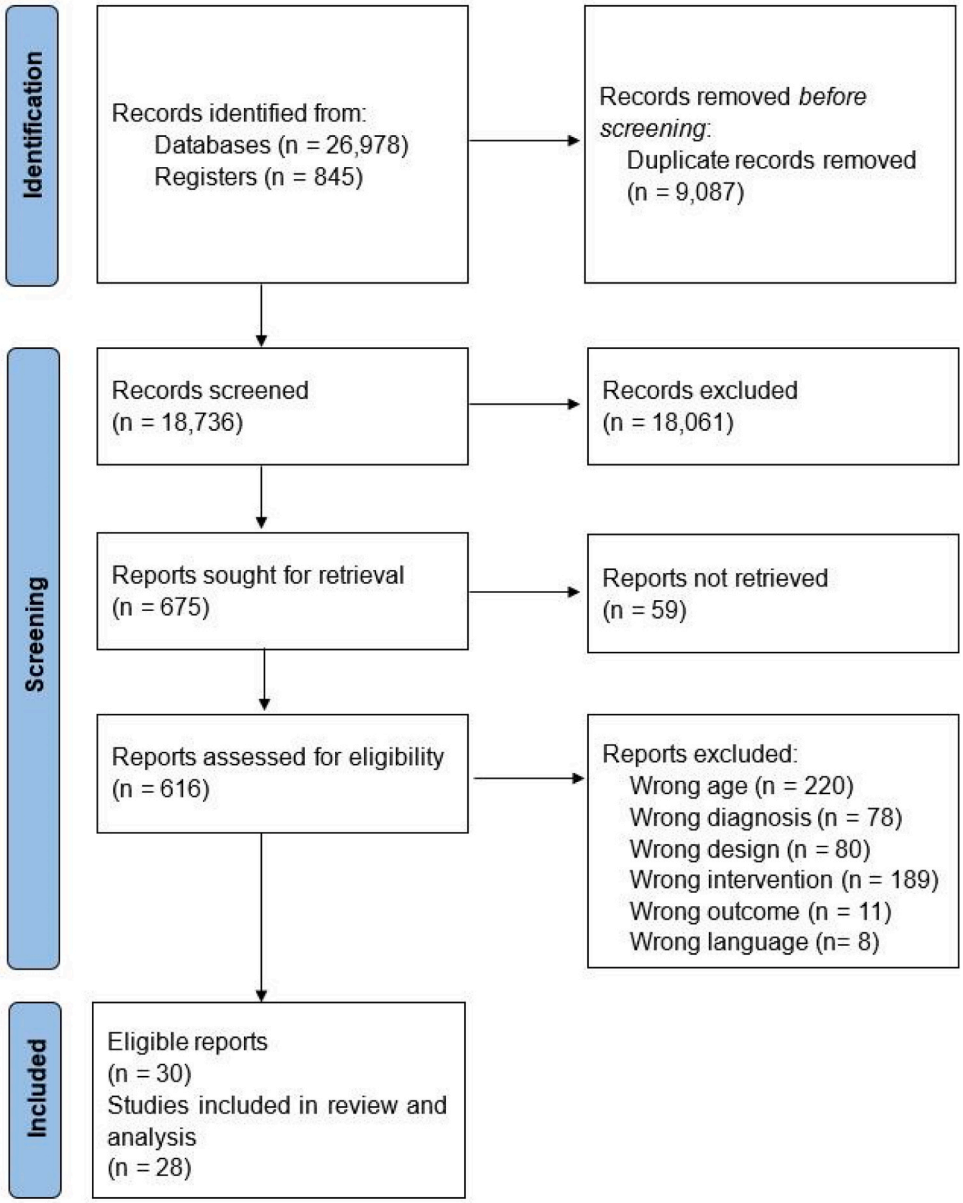


Fig. 1. Flow Chart.

for quality appraisal to measure the level of agreement.

5.4. Data analysis

We calculated Cohen's d with a 95% confident interval (CI) using raw data extracted from the included studies to assess the effect sizes. For studies that had more than one intervention arm, each intervention group was extracted with reference to a common control group. Where possible, we conducted robust variance estimation with 95% CIs to pool estimates for each type of intervention and each outcome domain. Such models allow us to include multiple estimates from each study and account for dependencies. The default value of 0.8 was used for within-study effect size correlation and small sample adjustment was incorporated to further reduce false positive errors (Fisher & Tipton, 2015; Tanner-Smith, Tipton, & Polanin, 2016). Outliers were defined as effect sizes for which the upper CI was smaller than the lower CI of the pooled effect or for which the lower CI was larger than the upper CI of the pooled effect (Viechtbauer & Cheung, 2010) and inspected prior to formal synthesis. Heterogeneity of studies was quantified using I^2 statistics and investigated in forest plots. We conducted sensitivity analyses to assess the impact of study quality by removing studies that were scored 6 or lower in the quality appraisal. Subgroup analyses were conducted to examine differences by diagnosis. Additional analyses were done to

Table 1
Results of Robust Variance Estimation (Post-Intervention).

		Language	Social and communication	Cognitive	Adaptive Living	Behavior	Motor
All	d	0.55	1.22	0.65	1.42	0.41	1.03
	95% CI	[0.33,0.76]	[0.68,1.76]	[0.14,1.17]	[− 0.38,3.22]	[− 0.74,1.57]	[− 0.10,2.16]
	I^2	77.88%	89.74%	87.23%	92.73%	86.21%	97.15%
	n(k)	20(81)	12(83)	11(31)	5(9)	4(17)	7(27)
Sensitivity Analysis	d	0.66	1.18	0.42	1.42	0.25	0.84
	95% CI	[0.09,1.22]	[0.5,1.86]	[− 0.15,0.98]	[− 0.38,3.22]	[− 5.82,6.32]	[− 4.17,5.86]
	I^2	74.47%	89.43%	67.67%	92.73%	88.43%	98.92%
	n(k)	7(17)	9(60)	4(13)	5(9)	2(4)	3(9)
CP	d	\	\	\	\	\	0.48
	95% CI						[− 2.73,3.69]
	I^2		\	\	\	\	68.59%
	n(k)	1(1)	1(1)	1(1)	1(1)	\	2(3)
DD	d	0.75	\	0.35	\		1.28
	95% CI	[− 1.09,2.59]		[− 0.74,1.43]			[0.02,2.53]
	I^2	90.88%	\	75.93%	\	\	77.28%
	n(k)	3(7)	\	3(8)		1(3)	2(16)
ID	d	\	\	\	\	\	\
	95% CI						
	I^2	\	\		\	\	\
	n(k)	\	1(17)	1(9)	\	1(10)	1(5)
LD	d	0.46	\	\	\		\
	95% CI	[0.2,0.72]					
	I^2	73.30%	\	\	\	\	\
	n(k)	11(65)	\	1(2)	\	\	\
Autism	d	0.79	1.02	0.44	1.33	0.25	0.34
	95% CI	[0.34,1.23]	[0.47,1.57]	[− 0.02,0.89]	[− 1.33,3.99]	[− 5.82,6.32]	[− 13.8,14.49]
	I^2	53.39%	88.73%	69.39%	93.42%	88.43%	96.14%
	n(k)	5(8)	10(65)	5(11)	4(8)	2(4)	2(3)

Note: CP-Cerebral Palsy, DD-Developmental Disabilities, ID-Intellectual Disability, LD-Language Disorder

investigate the effect of world region and country/region income level on pooled effect sizes. Publication bias was assessed for outcomes that were measured in at least 10 studies, by examining the asymmetry of funnel plots and quantifying the degree of asymmetry using Egger's test (Egger, Smith, Schneider & Minder, 1997). All data analyses were conducted in R (version 4.0.3) using the Robumeta package (Fisher & Tipton, 2015).

6. Results

The search resulted in 18,736 records after removing 9087 duplicates. A total of 18,061 studies were deemed ineligible for inclusion based on their titles and abstracts, which exhibited clear indications of non-conformance with the predefined inclusion criteria. Six hundred and sixteen full-texts were retrieved for further assessment, of which 586 were subsequently excluded primarily due to wrong intervention types, research designs, program participants, or languages (Fig. 1). Thirty study reports met the inclusion criteria. Two of them reported the same outcomes of an intervention at different time points and were therefore considered as one study (Kaale, Fagerland, Martinsen & Smith, 2014; Kaale, Smith, & Sponheim, 2012). Another study reported different outcomes of an intervention in two separate papers. We viewed the two papers as one study (Pears et al., 2016, 2015). In total, twenty-eight studies involving 2937 preschool children, published between 1999 and 2022, were included in the review and meta-analyses. There was a high level of reviewer agreement (unweighted kappa: 0.88, 95% CI [0.87, 0.90]; percentage: 99%). The PRISMA flowchart is presented in Fig. 1.

Nine (32%) studies involved children with autism; eight (29%) included children with language disorder; three (11%) were conducted among children with cerebral palsy; and two (7%) involved children with intellectual disability. There were also six (21%) studies including multiple types of developmental disabilities. Ten (36%) investigated the impacts of language interventions (e.g., shared book reading (Crain-Thoreson & Dale, 1999; Dawes et al., 2019; Van Kleeck et al., 2006)) for children with language disorders, autism, intellectual disability, and multiple types of developmental disabilities. A quarter (n = 7) tested social and communication skills training (e.g., Joint Attention Symbolic Play Engagement Regulation – JASPER (Chang et al., 2016; Goods et al., 2013)) for children with autism and intellectual disability. Three (11%) evaluated physical interventions (e.g., Young Athletes Program (Favazza, Siperstein, Zeisel, Odom, Sideris & Moskowitz, 2013)) for children with intellectual disability and multiple types of developmental disabilities. The remaining eight involved comprehensive school readiness programs (e.g., Kids in Transition to School; McDermott et al., 2018; Pears et al., 2016, 2015) that target multiple developmental domains for children with cerebral palsy, autism, and multiple types of developmental disabilities.

Sample sizes ranged from 15 to 312. All studies had an inactive control group, except for two. Observational measures and tests were used in all but one (n = 27, 96%) studies; professional (i.e., teacher or therapist) report in seven (25%) studies; and parent or caregiver report in four (14%) studies. Most (n = 23, 82%) studies were done in high-income countries/regions and five (18%) in a upper-middle-income country. Sixteen (57%) were from the Americas (all in the United States); seven (25%) from the Western Pacific

Table 2
Impact of Country Level Income and Geographical Region.

	Language	Social and communication	Cognitive	Adaptive Living	Behavior	Motor
Upper-Middle-Income	-0.38	0.58	1.26	-0.49	1.12	-1.31
	[− 2.27,1.52]	[− 1.25,2.40]	[− 0.86,3.38]	[− 5.44,4.47]	[− 0.85,3.09]	[− 3.55,0.92]
High-Income	0.59**	1.07**	0.33*	1.63	-0.10	1.59
	[0.35,0.82]	[0.45,1.70]	[0.04,0.63]	[− 3.24,6.51]	[− 1.35,1.15]	[− 0.13,3.31]
Europe	0.29	0.97	-0.32			
	[− 0.18,0.77]	[− 0.31,2.25]	[− 0.66,0.01]			
Western Pacific	0.25	1.00	1.07	-0.49	1.12	-1.31
	[− 0.62,1.13]	[− 0.59,2.58]	[− 0.15,2.29]	[− 5.44,4.47]	[− 0.85,3.09]	[− 3.55,0.92]
Americas	0.41*	0.65*	0.31	1.63	-0.10	1.59
	[0.12,0.69]	[0.12,1.19]	[− 0.02,0.65]	[− 3.24,6.51]	[− 1.35,1.15]	[− 0.13,3.31]

* note:, $p < 0.05$;
** $p < 0.01$

(China: n = 5, 18%; Hong Kong, China: n = 1, 4%; Australia: n = 1, 4%); and five (18%) from Europe (Norway: n = 2, 7%; Israel: n = 2, 7%; Germany: n = 1, 4%). Speech and language development was measured in 16 (62%) studies; cognitive development in 10 (36%) studies; social and communication skills in 10 (36%) studies; motor and physical development in six (21%) studies; behavioral adjustment in five (18%) studies; and adaptive living skills in three (11%) studies. Quality scores ranged from 1 to 10, with high reviewer agreement (weighted kappa: 0.93, 95% CI [0.87, 0.99]; percentage: 82%). The overall quality was medium on average (mean score = 5.5). Appendix Table 1 shows the characteristics of included studies.

6.1. *Synthesized findings, at post-intervention*

Four studies had two or three intervention arms (Bauminger-Zviely, Eytan, Hoshmand & Rajwan Ben, Shlomo, 2020; Crain-Thoreson & Dale, 1999; Lonigan & Phillips, 2016; Roden, Früchtenicht, Kreutz, Linderkamp & Grube, 2019). For those studies, each arm was compared to a common comparator and treated as a separate study in the meta-analytic models.

The results (Table 1) showed that the early childhood social and communication trainings delivered in preschool educational settings for children with autism, intellectual disability, and cerebral palsy might have large effects at post-intervention in promoting social and communication skills (Cohen's d = 1.22, 95% CI [0.68, 1.76]; 12 studies, 83 estimates; $I^2 = 89.74\%$). Programs targeting cognitive abilities among children with autism, intellectual disability, cerebral palsy, language disorder, and developmental disabilities in general might have medium effects in improving cognitive skills (d = 0.65, 95% CI [0.14, 1.17]; 11 studies, 31 estimates; $I^2 = 87.23\%$). There was also tentative evidence of interventions targeting language abilities in improving language skills in children with autism, intellectual disability, cerebral palsy, language disorder, and developmental disabilities in general (d = 0.55, 95% CI [0.33, 0.76]; 20 studies, 81 estimates; $I^2 = 77.88\%$). Comprehensive programs aiming to improve multiple developmental domains might have no statistically significant group difference at post-intervention in adaptive living skills among children with autism or cerebral palsy (d = 1.42, 95% CI [−0.38, 3.22]; 5 studies, 9 estimates; $I^2 = 92.73\%$) or child emotional and behavioral adjustment among those with autism, intellectual disability, or developmental disabilities (d = 0.41, 95% CI [−0.74, 1.57]; 4 studies, 17 estimates; $I^2 = 86.21\%$). The physical or movement programs might have no effect in promoting physical and motor development for children with autism, intellectual disability, cerebral palsy, or developmental disabilities (d = 1.03, 95% CI [−0.10, 2.16]; 7 studies, 27 estimates; $I^2 = 97.15\%$). After removing studies with low quality scores, the results remained consistent for social and communication skills (d = 1.18, 95% CI [0.50, 1.86]; 9 studies, 60 estimates; $I^2 = 89.43\%$) and language development (d = 0.66, 95% CI [0.09, 1.22]; 7 studies, 17 estimates; $I^2 = 74.47\%$). However, the effect on cognitive development (d = 0.42, 95% CI [−0.15, 0.98]; 4 studies, 13 estimates; $I^2 = 67.67\%$) was not maintained. There was substantial heterogeneity across outcomes. All forest plots are presented in the appendices.

Publication bias was assessed for language development, social and communication skills, and cognitive development at post-intervention. All funnel plots were asymmetrical, with a significant Egger's test (social and communication skills: $b = 0.77$, $p < .000$; language skills: $b = 0.42$, $p < .000$; cognitive development: $b = 0.35$, $p = .002$). However, concerning language skills, the funnel plot revealed a greater number of studies in areas of non-significance compared to areas of higher significance. This suggests the presence of other factors, such as genuine heterogeneity or suboptimal methodological design, that may be contributing to the observed asymmetry (Sterne et al., 2011). All funnel plots are presented in the appendices.

The results of subgroup analyses provided tentative evidence that the early childhood language interventions might improve language skills in children with language disorder (d = 0.46, 95% CI [0.20, 0.72]; 11 studies, 65 estimates; $I^2 = 73.30\%$) and in those with autism (d = 0.79, 95% CI [0.34, 1.23]; 5 studies, 8 estimates; $I^2 = 53.39\%$) at post-intervention. Results also showed a potentially large effect at post-intervention on social and communication skills for children with autism (d = 1.02, 95% CI [0.42, 1.57]; 10 studies, 65 estimates; $I^2 = 88.73\%$). The results of data from two studies showed some evidence of physical interventions in improving motor skills in children with multiple types of developmental disabilities (d = 1.28, 95% CI [0.02,2.53]; 2 studies, 16 estimates; $I^2 = 77.28\%$) at post-intervention. The subgroup analyses identified no evidence of effects for other outcomes domains or for children with cerebral palsy or intellectual disability. Heterogeneity ranged from moderate (46.12%) to substantial (99.09%).

Meta-regressions (Table 2) showed that compared to interventions implemented in upper-middle-income countries, programs conducted in high-income countries/regions might have a stronger effect at post-intervention in terms of improving language (b =

Table 3
Results of Robust Variance Estimation (Follow-Up).

	Language	Social and communication	Cognitive	Adaptive Living	Behavior	Motor
d 95% CI	0.27 [− 0.40,0.94]	0.49 [− 2.86,3.85]			0.61 [−4.15,5.36]	
I^2	71.33%	53.22%			80.65%	
n(k)	4(16)	2(10)	1(2)	1(2)	2(2)	1(4)

0.59, 95% CI [0.35, 0.82]), social and communication ($b = 1.07$, 95% CI [0.45, 1.70]), and cognitive skills ($b = 0.33$, 95% CI [0.04, 0.63]). Programs delivered in the Americas (United States) tended to be more likely to promote language development ($b = 0.41$, 95% CI [0.12, 0.69]) and social and communication skills ($b = 0.65$, 95% CI [0.12, 1.19]) in preschool children with developmental disabilities than in other regions.

6.2. Synthesized findings, at follow-up

Follow-up assessments were conducted in five studies (Dawes et al., 2019; Hagen, Melby-Lervåg, & Lervåg, 2017; pp. 2, 1276; Kaale et al., 2014; Pears et al., 2015; Zeng et al., 2021). Follow-up time points varied substantially, ranging from 8 weeks to 1 year. Exploratory analyses (Table 3) provided no evidence of an long-term impact on social and communication skills (d = 0.49, 95% CI [−2.86, 3.85]; 2 studies, 10 estimates; $I^2 = 53.22\%$), language development (d = 0.27, 95% CI [−0.40, 0.94]; 4 studies, 16 estimates; $I^2 = 71.33\%$; forest plot), or emotional and behavioral adjustment (d = 0.61, 95% CI [−4.15, 5.36]; 2 studies, 2 estimates; $I^2 = 80.65\%$; forest plot) for children with language disorder, autism, or multiple types of developmental disabilities. There was moderate to substantial heterogeneity among studies.

7. Discussion

7.1. Main findings

This review aims to assess the effectiveness of early childhood interventions delivered in educational settings for promoting school readiness for children aged 3–6 years with autism and other developmental disabilities. A total of 28 randomized controlled trials involving 2937 preschool children with five types of developmental disabilities were identified, all of which included post-intervention data. Follow-up data were provided in five of the trials.

Our results suggest that there is inconclusive evidence of the effectiveness of early childhood social and communication trainings in enhancing the social and communication skills of children with autism, intellectual disability, and cerebral palsy at post-intervention (d = 1.22, 95% CI [0.68, 1.76]). The sensitivity analysis suggested a high level of robustness. The finding is in line with the disability literature that indicates the efficacy of early education programs, such as social skills training, dialogic reading, and phonological awareness training, in promoting communication and social development for children with disabilities or developmental delays in the United States (What Works Clearinghouse, 2010a, 2012, 2013).

The potentially large effect size found in our review might be caused by publication bias. Additionally, studies conducted in low-resource settings, such as China (Chen, Li, Tang & Liu, 2013), where early childhood services for children with developmental disabilities are scarce, could have resulted in more significant changes. Our review revealed a stronger effect in children with autism (d = 1.02, 95% CI [0.42, 1.57]), consistent with another review showing that communication interventions improved spoken and non-verbal communication in children with autism (Brignell, Chenausky, Song, Zhu, Suo & Morgan, 2018). However, the small numbers of studies for other diagnoses, with only one study for cerebral palsy, one for intellectual disability, and none for language disorder, may also account for the lack of effects.

We also found provisional evidence of early childhood language interventions for children with autism, intellectual disability, cerebral palsy, language disorder, and developmental disabilities in general, in promoting language skills at post-intervention (d = 0.55, 95% CI [0.33, 0.76]). The treatment effect was maintained in the sensitivity analyses, suggesting a high degree of certainty. This is consistent with previous reviews that have shown positive effects of school readiness programs on language competencies in children with developmental delay or disabilities in the United States (What Works Clearinghouse, 2010a, 2012). Our results further suggest that interventions for children with autism (d = 0.79, 95% CI [0.34, 1.23]) and language disorder (d = 0.46, 95% CI [0.20, 0.72]) might have a stronger effect than those for other diagnoses. These findings are in line with existing systematic reviews demonstrating positive effects of early programs on language development for children with language disorders, including stuttering reduction (Law, Garrett, & Nye, 2003) and expressive and receptive language improvement for children with autism (Reichow, Hume, Barton & Boyd, 2018). Additionally, early childhood programs and school interventions have shown potential for promoting language development in children with intellectual disability (Reichow, Lemons, Maggin & Hill, 2019). However, our review could not confirm this due to the limited number of studies available.

Our finding suggests that early childhood interventions targeting cognitive abilities in preschool settings might have improved cognitive skills among children with autism, intellectual disability, cerebral palsy, language disorder, and developmental disabilities (d = 0.65, 95% CI [0.14, 1.17]) at post-intervention but such effect was not maintained after removing studies with low quality scores. The result is similar to an existing review on early intensive interventions for children with autism, which also identified no evidence of effectiveness in child intellectual ability (Reichow et al., 2018). In contrast, a meta-analysis of early education for children without

disabilities found significant improvements in cognitive outcomes (Camilli et al., 2010). The reason may be twofold. Among studies that contributed to our pooling of effects on cognitive development, only one considered cognitive development as the primary focus, with the remaining studies targeting more than one developmental domain. Cognitive impairment in children with intellectual disability, cerebral palsy, and autism might have also posed additional challenges. However, given missing information on child comorbidities, we were not able to further investigate the impact of such impairment.

We detected no effect of movement interventions in promoting physical well-being in children with autism, intellectual disability, cerebral palsy, or developmental disabilities (d = 1.03, 95% CI [−0.10, 2.16]). This is line with a systematic review that synthesized the effects of interventions to promote movement skills in children with developmental coordination disorder aged 5–12 years, finding no significant group difference among the most robust studies (Miyahara, Hillier, Pridham & Nakagawa, 2017). However, it is important to acknowledge that in both reviews, the number of identified studies was insufficient to draw definitive conclusions regarding the effectiveness of interventions on motor skills. Additionally, despite a narrower age range, our sample exhibited greater diversity in terms of disability type and contextual factors, which may have contributed to increased heterogeneity and decreased the precision of the finding.

Similarly, our review found no evidence of comprehensive early childhood programs on adaptive living skills among children with autism or cerebral palsy (d = 1.42, 95% CI [−0.38, 3.22]) or emotional and behavioral adjustment among children with autism, intellectual disability, or developmental disabilities (d = 0.41, 95% CI [−0.74, 1.57]) at post-intervention. The finding differs from evidence found in the U.S., suggesting that social skills training in early educational settings has positive effects on emotional development and behaviors for children with special needs (What Works Clearinghouse, 2013). However, it should be noted that our analysis included studies conducted in three different countries and more comprehensive school readiness programs in addition to social skills training. The limited numbers of studies in our meta-analytic models also require caution in interpreting the results.

Our analysis did not find statistically significant group differences for any of the outcome domains in the follow-up assessments. However, caution is needed when interpreting these results, as the number of studies conducting follow-ups was limited and the follow-up time points varied. Nonetheless, our exploratory findings might speak to previous reviews of early interventions for children with and without developmental disabilities, which have either reported no follow-up data (Pennington, Akor, Laws & Goldbart, 2018; Sjøstrand et al., 2021) or the fadeout of program effects 12 months after the completion of intervention, despite the observation of benefits in adolescence and adulthood (Bailey et al., 2016; O'toole, Lee, Gibbon, van Bysterveldt & Hart, 2018).

While prior research has established that the greatest benefits of early childhood programs accrue to children from the most disadvantaged contexts (Arnold et al., 2007), our meta-regressions indicated that, in general, early childhood programmes for preschool children with developmental disabilities implemented in high-income countries or regions and in North America might be more effective in improving language, social, and communication skills than those in the upper-middle-income country and other world regions. One potential explanation is that programs in high-income settings have greater access to resources such as well-trained staff and better facilities, which may improve their adherence to program protocols and enhance treatment effects (Durlak & DuPre, 2008; Furlong, McGilloway, Bywater, Hutchings, Smith & Donnelly, 2012). However, less than half (n = 11, 39%) of our included studies conducted fidelity measures and we were therefore unable to test this assumption. Another potential explanation is that the programs implemented in the upper-middle-income country predominantly consisted of homegrown interventions, whose evidence base remains uncertain. Some of these interventions also encompassed a broad range of components targeting multiple developmental domains, which enhanced their comprehensiveness but potentially compromised the depth and impact of the intervention. In contrast, the interventions identified in high-income countries were more likely to be well-established programs that were developed based on established theories and empirical evidence, thereby increasing their likelihood of achieving desired outcomes.

7.2. Strengths and limitations

This review has multiple strengths. It used comprehensive search terms to identify relevant studies in international and regional databases, which was further supplemented with grey literature repositories and hand-searching of reference lists. The search of Chinese reports contributed 14% of the included studies and 80% of the studies conducted in low- and middle-income countries, underscoring the importance of seeking evidence using languages other than English. The review process involved two authors working independently at each stage, which helped minimize errors. High levels of agreement were achieved during screening and quality appraisal. This indicated that the reviewers interpreted and applied the criteria in a consistent manner, which ensured the reliability of the review. The data syntheses were conducted using robust variance estimation, which allowed for the incorporation of clustering in the meta-analysis, resulting in more accurate standard errors and confidence intervals. This increased the number of estimates included in each model and prevented second-order sampling bias. Robust variance estimation also helped to account for heterogeneity and improved the accuracy of the overall effect size estimates. Additionally, all but one study used observational measures, providing data less affected by biases from self-report, such as social desirability bias and recall bias.

This review also exhibits several limitations. A major source of bias is the substantial heterogeneity among studies, caused by differences in program content and features, methodological designs, deliverers, participant characteristics (e.g., types of disabilities and potential comorbidities that we were not able to rule out), and intervention contexts. The results were thus inconclusive and should be treated with caution. Another limitation is the small number of included studies, which underpowered the analyses, in particular for adaptive living skills, behavioral adjustment, and motor development, as well as for the subgroup analyses and meta-regressions. Also, we did not identify studies involving children with other developmental disabilities, such as ADHD, epilepsy, and hearing loss. While there is some evidence advising against diagnosing ADHD at an early age (Greenhill, Posner, Vaughan & Kratochvil, 2008), other developmental disabilities can be diagnosed prior to entering school. Research from high-income countries and

regions dominates, with three fourths of the studies from the U.S. and European countries. Evidence is missing from low- and middle-income countries other than China. No eligible studies were found from areas such as Latin America, Central and South/-Southeast Asia, Africa, and Eastern Mediterranean. Quality of included studies was rated medium on average, which further reduced our confidence in the findings. Only five studies conducted follow-ups, limiting our ability to assess the sustainability of program effects.

7.3. Implications

The exclusion of children with autism and other developmental disabilities from early childhood services, particularly in low-resource settings, can lead to increased social inequality and health disparities. Our review highlights the potential benefits of providing early childhood interventions in educational settings to promote the development and school readiness for young children with autism and other developmental disabilities aged 3–6. We call for intersectional partnerships in strengthening existing evidence and conducting more research, particularly in low- and middle-income countries, to investigate the impacts of such intervention. International collaborations can also be established to increase funding and guidance to support capability building and program implementation in low-resource settings. Creating an enabling environment is also crucial for promoting school readiness as it is shaped by the interactions between children and their surroundings. Therefore, it is imperative to establish a supportive ecosystem by also creating "ready schools" and "ready families". Additionally, broad-based campaigns that reduce stigma and discrimination against children with autism and other developmental disabilities, as well as their families, is necessary to foster an inclusive service and education system.

Future research should focus on strengthening the methodological rigor, reaching children with more types of developmental disabilities and examining variations in treatment effects. Implementation evidence, such as quality of delivery, methods of delivery in emergency contexts, balance between fidelity and adaption, and participant engagement, should also be documented. Extended time points of assessments are recommended to assess the long-term impact of such interventions. Future research should also report cost and cost-effectiveness and investigate the optimization of interventions, to ensure the best use of limited funds.

CRediT authorship contribution statement

ZF and DQ conceptualized and designed the study. CZ and ZF contributed to the screening. XL and ZF extracted the data and conducted quality appraisal. ZF analyzed the data and drafted the manuscript. All authors contributed to manuscript revision and approved the submission.

Declaration of Competing Interest

We declare no conflict of interest.

Data Availability

Data will be made available on request.

Appendices 1

Study Characteristics Table.

ID	Country	Sample Size	Mean Age	Diagnosis	Intervention	Length, Number of Sessions, Frequency	Outcome	Quality Score
Apache 2005 (80)	US	28	4.2	DD	Activity-based physical intervention - motor skill training	Length: 15 weeks Intended Sessions: 45 Frequency: 3 times per week	Test of Gross Motor Development	6
Bauminger-Zviely 2020 (62)	Israel	65	4.36–4.88	ASD	Arm 1: peer social intervention - social interaction Arm 2: peer social intervention - social conversation	Length: 6 months Intended Sessions: 78 Frequency: 3	Conversation Quality scale Peer Play Scale Pragmatics Rating Scale-Youth Social Play Questionnaire Social Conversation	9

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ID	Country	Sample Size	Mean Age	Diagnosis	Intervention	Length, Number of Sessions, Frequency	Outcome	Quality Score
					Arm 3: peer social intervention - social-pretend play	times per week	Questionnaire Vineland Adaptive Behavior Scale	
Boyd 2018 (81)	US	161	4.1	ASD	Advancing social-communication and play (ASAP) intervention	Length: 1 year Intended Sessions: unclear Frequency: daily	Autism Diagnostic Observation Schedule–Generic Caregiver-Teacher Rating Form Classroom engagement Structured Play Assessment	10
Chang 2016 (31)	US	66	4.19	ASD	Joint Attention Symbolic Play Engagement Regulation (JASPER)	Length: 8 weeks conducted over 5 months Intended Sessions: unclear Frequency: twice per day	Early Social-Communication Scales Structured play assessment Teacher-child play interactions	9
Chen 2013 (70)	China	30	4.38	ID	Music activities and play sessions	Length: 20 weeks Intended Sessions: 100 Frequency: 5 times per week	Cognitive abilities Emotional and behavioral adjustment Communicational skills Social skills	2
Crain-Thoreson 1999 (37)	US	37	4.3	LD	ARM 1: parent mediated one-on-one shared book reading practice ARM 2: staff led one-on-one shared book reading practice	Length: 8 weeks Intended Sessions: unclear Frequency: unclear	Child Language Data Exchange System Expressive One-Word Picture Vocabulary Test-Revised Peabody Picture Vocabulary Test-Revised Utterance	7
Dawes 2019 (38)	Australia	37	5	LD	Inferential comprehension intervention - shared book reading	Length: 8 weeks Intended Sessions: 16 Frequency: twice per week	Peter and the Cat Narrative Comprehension Assessment Squirrel Story Narrative Comprehension Assessment	3
Favazza 2013 (60)	US	233	4	ID	Young Athletes program - motor skill training	Length: 8 weeks Intended Sessions: 24 Frequency: 3 times per week	Peabody Developmental Motor Scales—Second Edition Vineland II, Teacher Rating Form	7
Goods 2013 (33)	US	15	4.325	ASD	Joint Attention Symbolic Play Engagement Regulation (JASPER)	Length: 12 weeks Intended Sessions: 24 Frequency: twice per week	Reynell Developmental Language Scales Structural Play Assessment Early Social Communication Scales Classroom Observation Measure	3
Goodway 2003 (82)	US	63	4.9–5.0	DD	Successful Kinesthetic Instruction for Preschoolers - motor skill training	Length: 9 weeks Intended Sessions: 18 Frequency: 5 days per week	Test of Gross Motor Development	6
Hagen 2017 (65)	Norway	301	4.82	LD	Language program consisting of dialogic reading and direct instructions	Length: 30 weeks Intended Sessions: 90 Frequency: 3 times per week	Illinois Test of Psycholinguistic Abilities British Picture Vocabulary Scale II Test for Reception of Grammar, Version II Vocabulary tests	5

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ID	Country	Sample Size	Mean Age	Diagnosis	Intervention	Length, Number of Sessions, Frequency	Outcome	Quality Score
							Wechsler Preschool and Primary Scale of Intelligence-III	
Han 2018 (83)	China	80	4.79–4.73	CP	Comprehensive training program targeting multiple developmental domains	Length: 3 months Intended Sessions: about 60 Frequency: 5 times per day	Gesell Developmental Schedules	7
Kaale 2012, 2014 (58,59)	Norway	61	3.97–4.19	ASD	Joint attention intervention	Length: 8 weeks Intended Sessions: 80 Frequency: 5 days per week	Early Social-Communication Scales Reynell Developmental Language Scales Social Communication Questionnaire	9
Keung 2022 (84)	Hong Kong SAR, China	68	4.83	DD	Think Bright intervention - cognitive and language intervention	Length: 6 months Intended Sessions: 12 Frequency: once every other week	Comprehensive Assessment Scales for Preschool Children Wechsler Intelligence Scale for Children Wechsler Preschool and Primary Scale of Intelligence-Fourth Edition	7
Lawton 2012 (32)	US	16	3.58–3.83	ASD	Joint Attention Symbolic Play Engagement Regulation (JASPER)	Length: 6 weeks Intended Sessions: 11 Frequency: 5 days per week	Class Observation Early Social-Communication Scales Taped Play Interaction	9
Lonigan 2016 (63)	US	93	4.85	LD	ARM 1: in-depth, teacher-directed, small-group instruction in phonological awareness, print knowledge, and oral language. ARM 2: narrower in scope, more intensive, and delivered to smaller groups of children.	Length: 11 weeks Intended Sessions: 44–55 Frequency: 5 days per week	Clinical Evaluation of Language Fundamentals–Preschool Code and Language Intervention Post-test Preschool Comprehensive Test of Phonological and Print Processing Test of Early Reading Ability-Third Edition	4
McDermott 2018 (36)	US	41	5.24–6.26	DD	Kids in Transition to School - school readiness intervention	Length: 4 months Intended Sessions: 24 Frequency: 2 times per week followed by once per week	Cognitive abilities	4
Pears 2015, 2016 (34,35)	US	225	5.26–5.28	DD	Kids in Transition to School Program - school readiness intervention	Length: 4 months Intended Sessions: 24 Frequency: 2 times per week followed by once per week	Behavior Rating Inventory of Executive Function – Preschool Parent Report Child Behavior Checklist Concepts About Print Emotion Regulation Checklist Dynamic Indicators of Basic Early Literacy Skills	4
Prince 2010 (85)	US	41	3.75–4	ASD, ID	Word Wall Intervention	Length: 3 months Intended Sessions: about 60 Frequency: 5 days per week	Diagnostic Assessment of Reading	5

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ID	Country	Sample Size	Mean Age	Diagnosis	Intervention	Length, Number of Sessions, Frequency	Outcome	Quality Score
Roden 2019 (61)	Germany	101	4.52	LD	Arm 1: auditory stimulation training Arm 2: pedagogical activity group	Length: 12 weeks Intended Sessions: 36 Frequency: 3 times per week	Auditory Screening Test Phoneme Discrimination Test Speech Perception Test	1
Strain 2011 (86)	US	294	4.175–4.225	ASD	Learning Experiences and Alternative Program	Length: 2 years Intended Sessions: unclear Frequency: unclear	Childhood Autism Rating Scale Mullen Scales of Early Learning Preschool Language Scale Social Skills Rating System	6
Tzuriel 2017 (87)	Israel	94	unclear	LD	Classification training	Length: 2 months Intended Sessions: 16 Frequency: twice per week	The Children's Conceptual and Perceptual Analogical Modifiability Test—Closed Analogies Version Katzenberger Assessment of Language in Preschool Age Language tests	2
van Kleeck 2006 (39)	US	30	4.17	LD	Shared book reading	Length: 8 weeks Intended Sessions: 16 Frequency: twice per week	Peabody Picture Vocabulary Test-III Preschool Language Assessment Instrument	4
Wang 2015 (88)	China	40	4.16–4.35	CP	Comprehensive training program targeting multiple developmental domains	Length: 6 months Intended Sessions: about 240 Frequency: 5 days per week	Gross Motor Function Measure Functional Independence Measure for Children	2
Wei 2013 (89)	China	56	4.4–4.5	CP	Comprehensive training program targeting multiple developmental domains	Length: 6 months Intended Sessions: about 240 Frequency: 5 days per week	Functional Independence Measure for Children	3
Wilcox 2020 (90)	US	289	4.24	LD	Teaching early literacy and language curriculum	Length: 34 weeks Intended Sessions: about 170 Frequency: 5 days per week	Clinical Evaluation of Language Fundamentals Preschool 2nd Edition Phonological Awareness and Literacy Screening Preschool Test of Preschool Early Literacy	4
Young 2016 (90)	US	312	4.31–4.50	ASD	Comprehensive Autism Program	Length: 1 school year Intended Sessions: unclear Frequency: daily	Autism Screening Instrument for Educational Planning, 3rd Edition Battelle Developmental Inventory Expressive One-Word Vocabulary Test Social Skills Rating Scale Vineland Adaptive Behavior Scales	6
Zeng 2021 (66)	China	60	4.54–4.55	ASD	Treatment and Education of Autistic and Communication Handicapped Children (TEACCH) & Discrete Trial Teaching (DTT)	Length: 6 months Intended Sessions: about 240 Frequency: 5 days per week	Chinese version of the Psycho-educational Profile, 3rd Edition	10

Note: ASD-Autism Spectrum Disorder, CP-Cerebral Palsy, DD-Developmental Disabilities, ID-Intellectual Disability, LD-Language Disorders

Appendices 2

Note: We conducted robust variance estimation with 95% CIs to pool estimates for each type of intervention and each outcome domain. The estimate numbers in the forest plots represent the ID of each estimate.

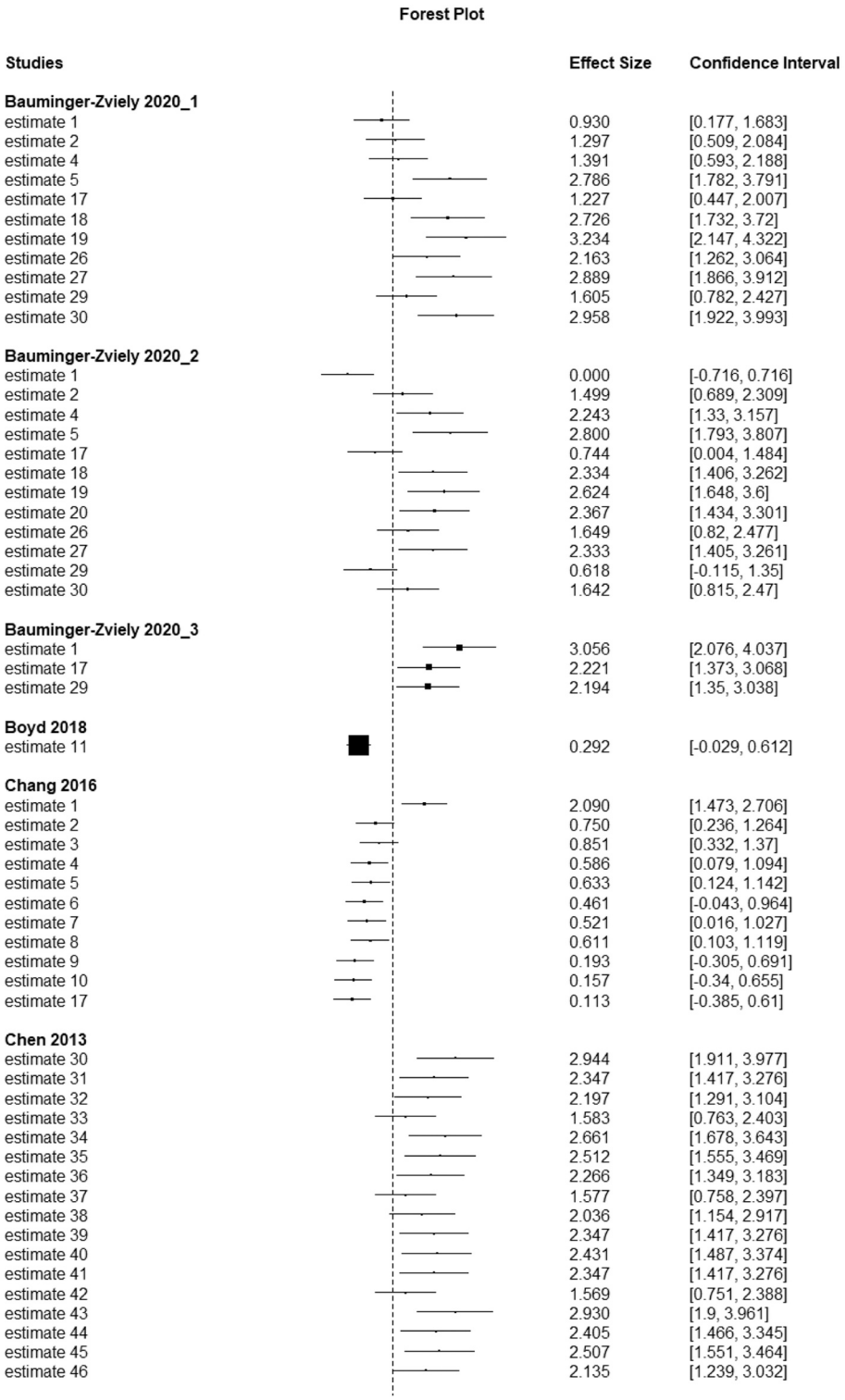


Fig. 1. Forest Plot, Social and Communicational Skills, at post-intervention.

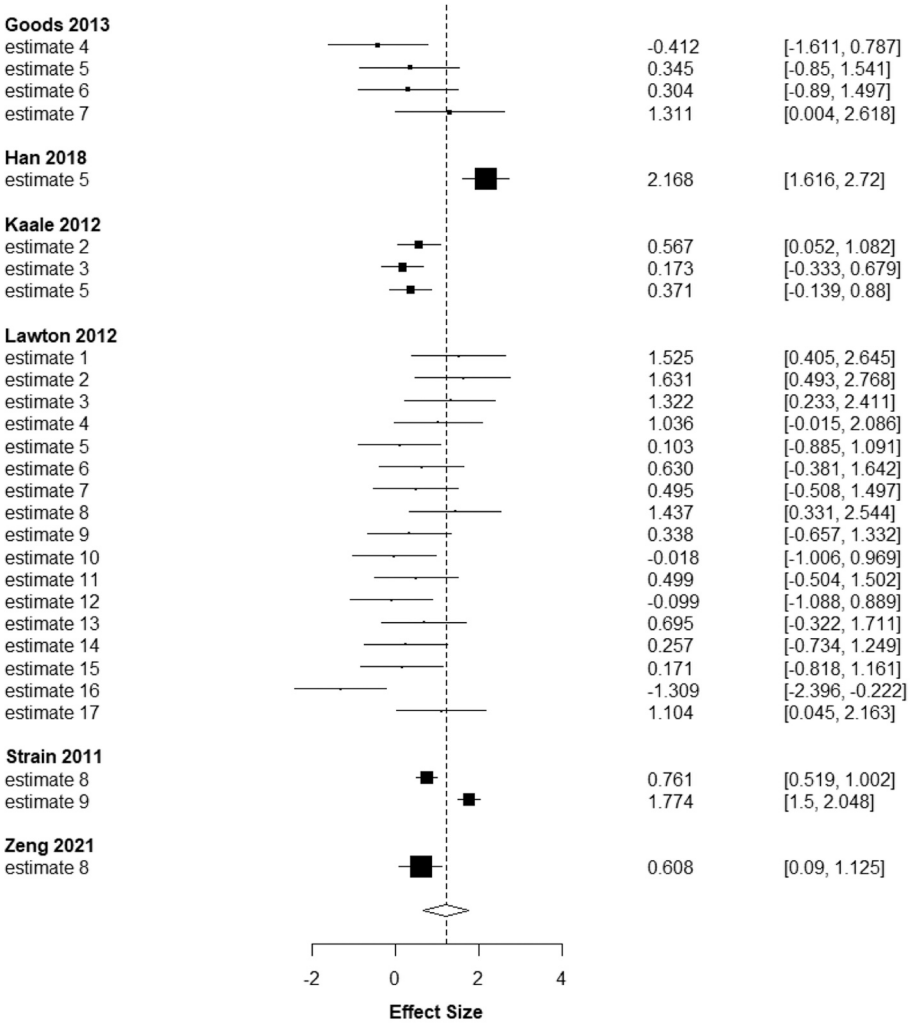


Fig. 1. (*continued*).

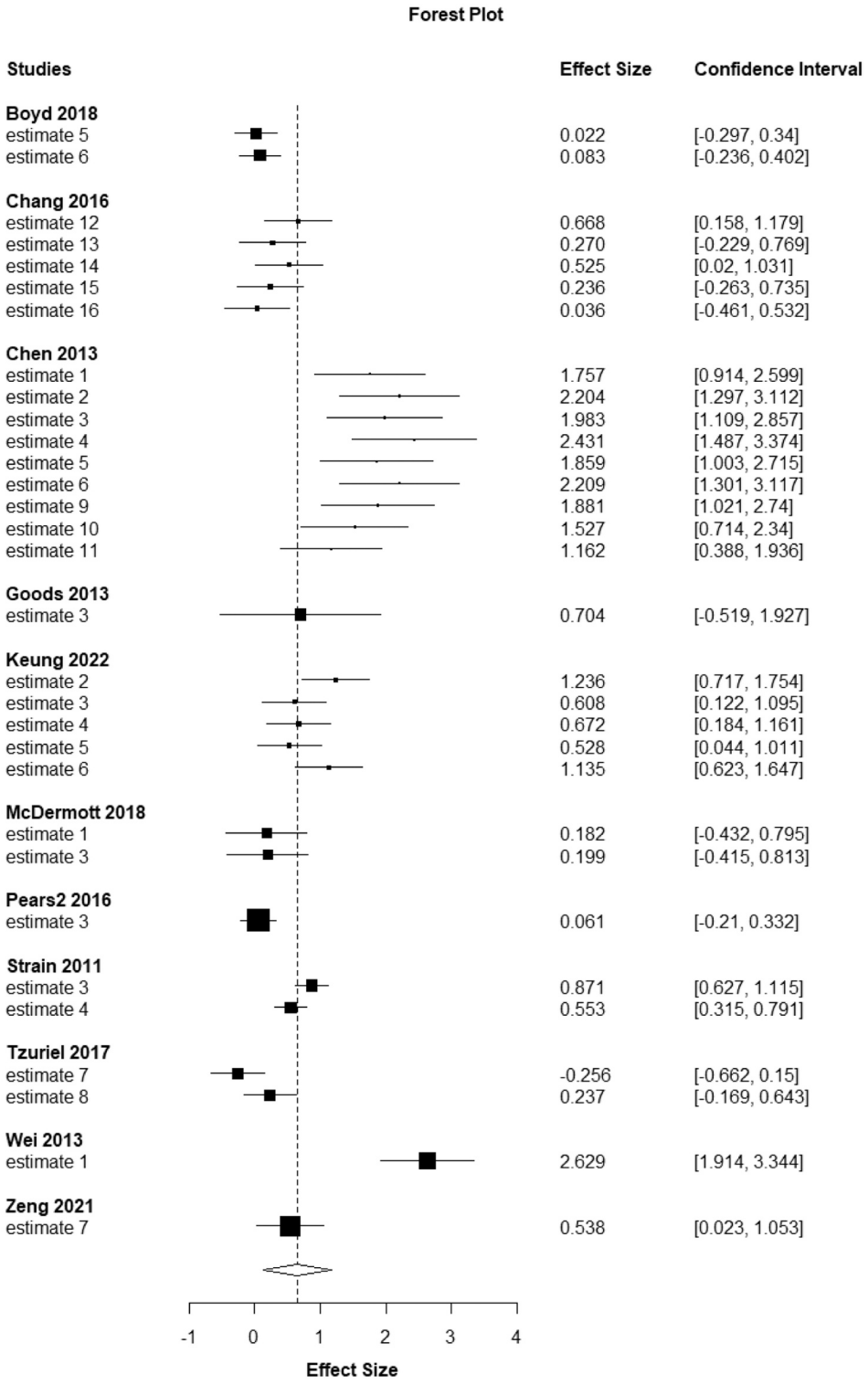


Fig. 2. Forest Plot, Cognitive Skills, at post-intervention.

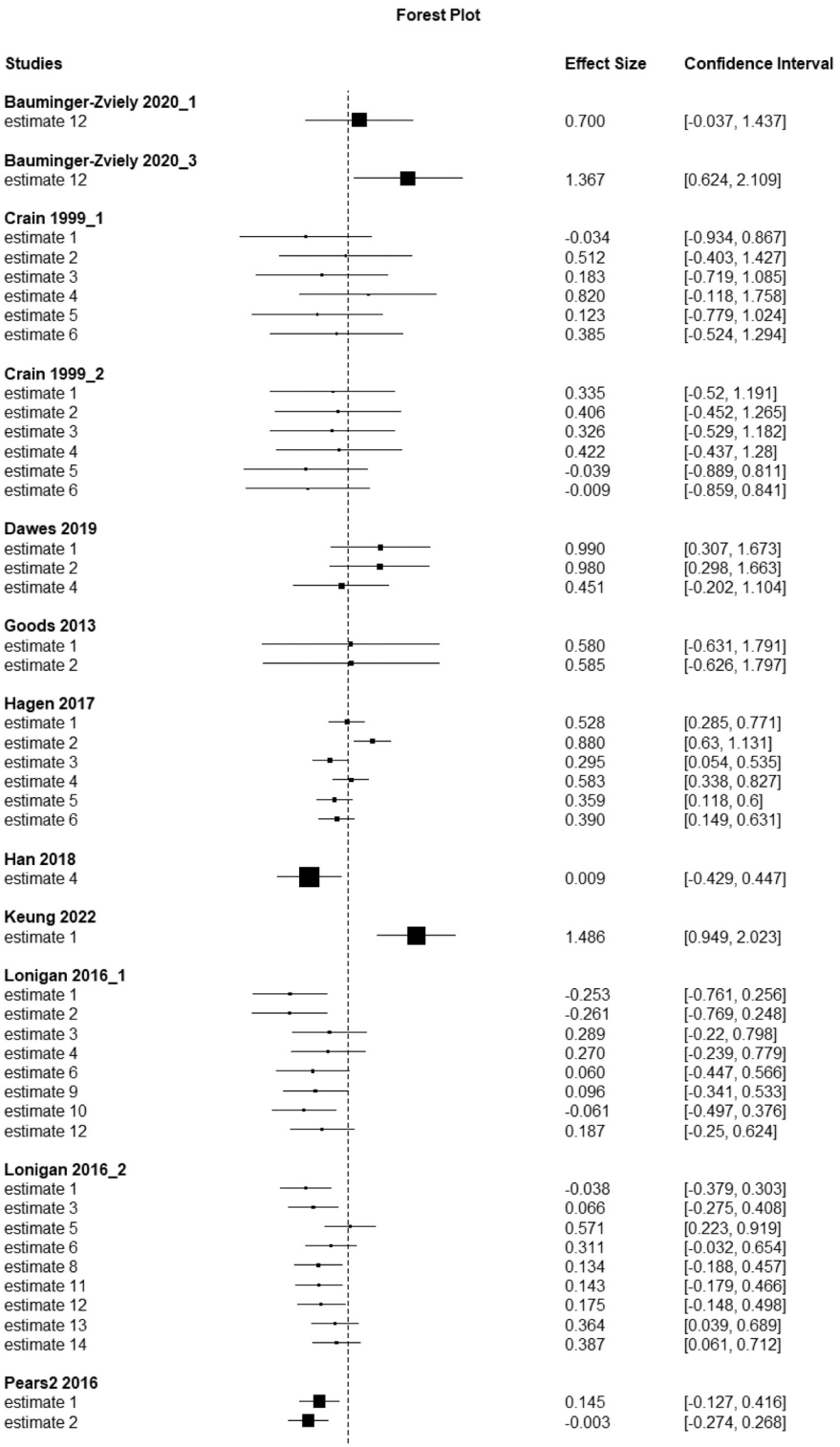


Fig. 3. Forest Plot, Language Skills, at post-intervention.

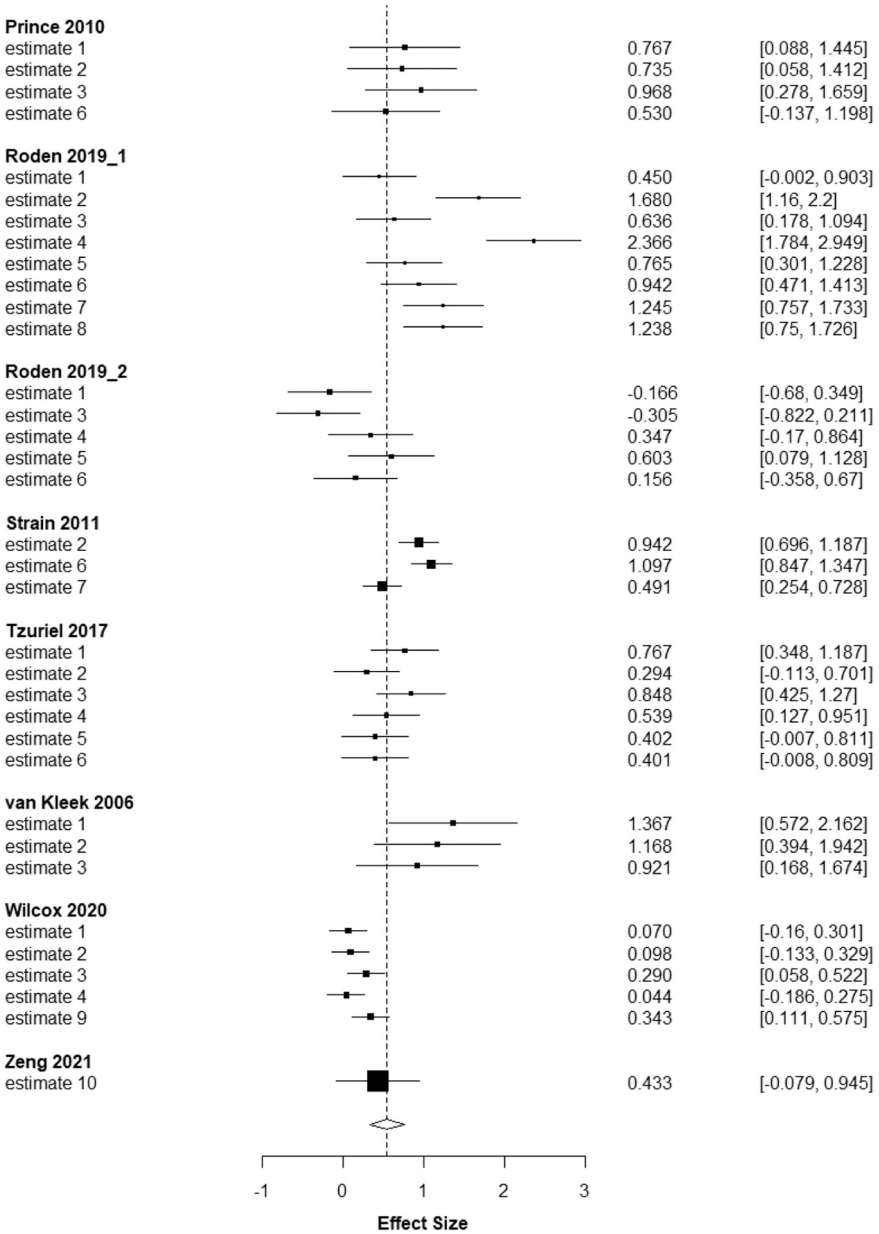


Fig. 3. (*continued*).

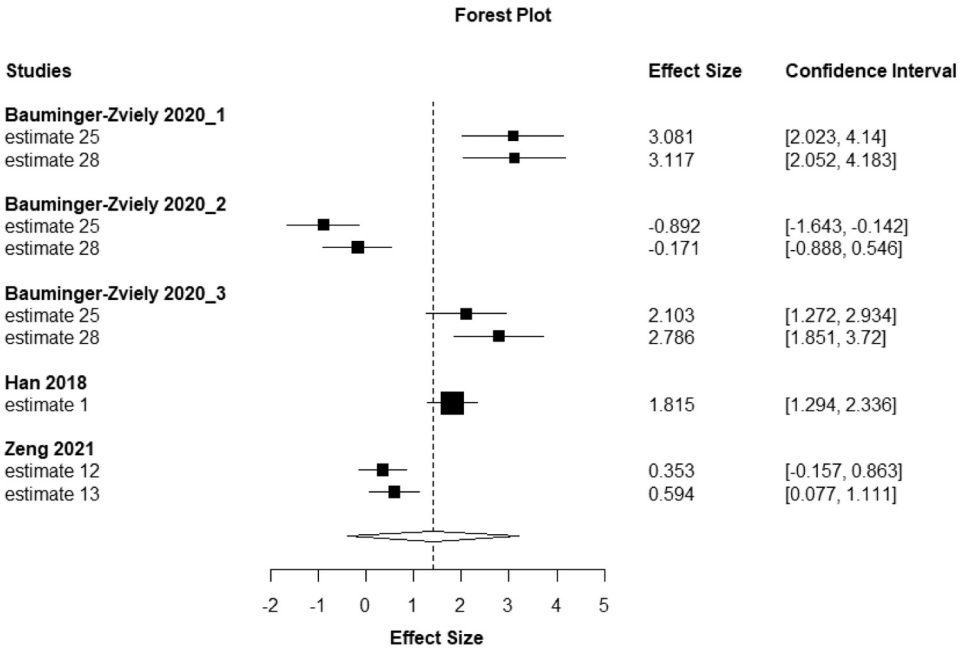


Fig. 4. Forest Plot, Adaptive Living Skills, at post-intervention.

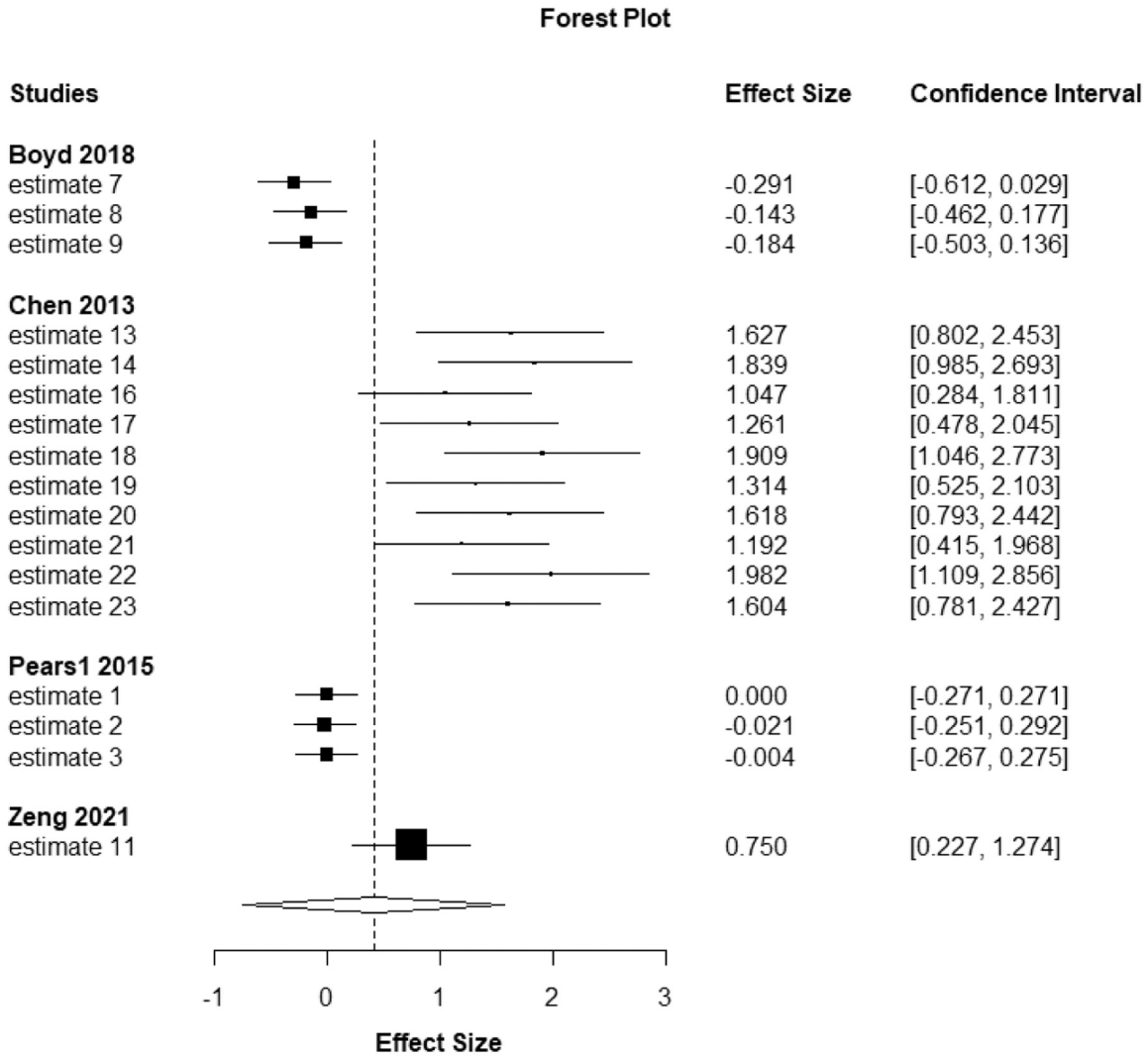


Fig. 5. Forest Plot, Emotional and Behavioral Adjustment, at post-intervention.

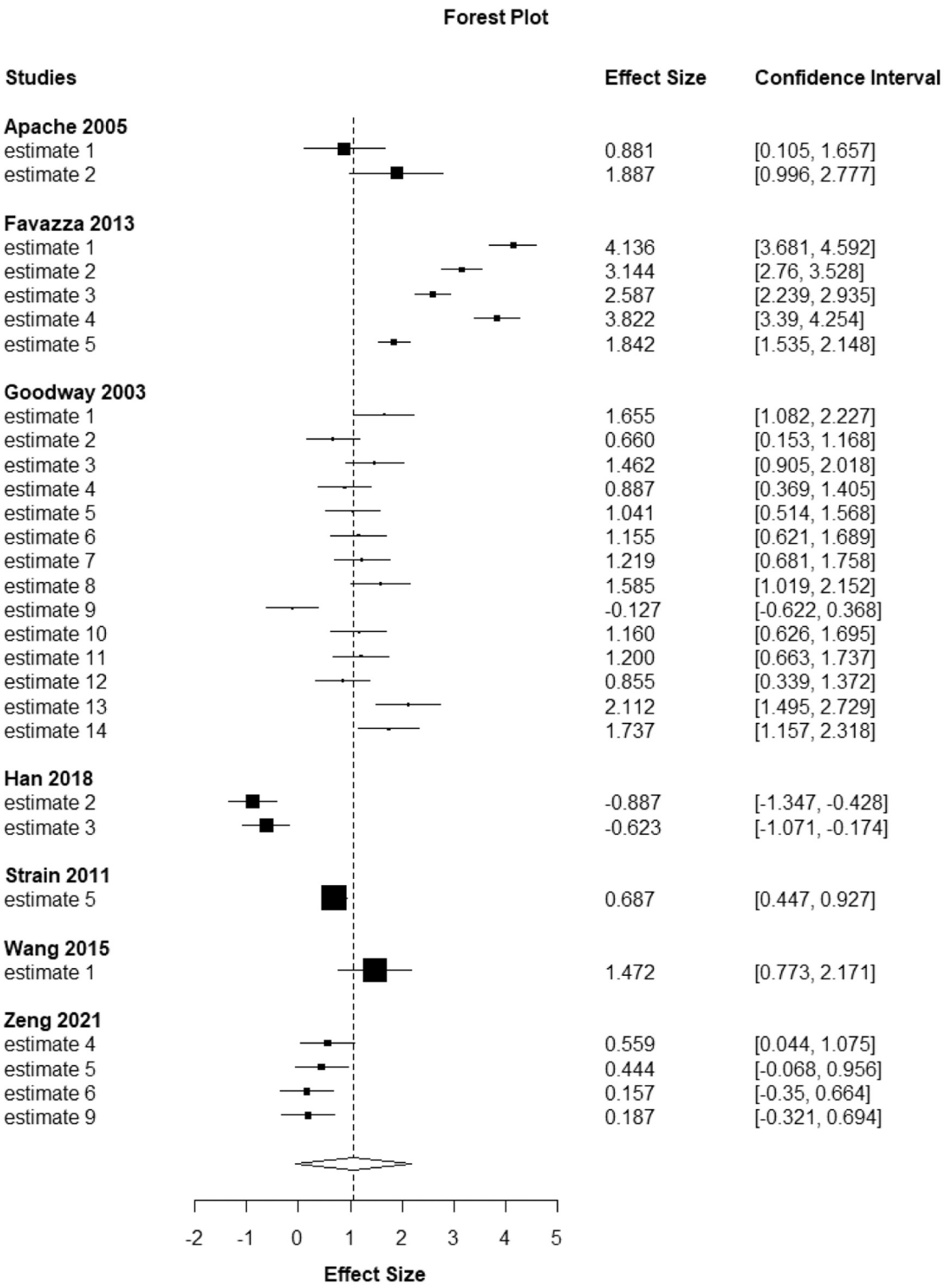


Fig. 6. Forest Plot, Physical and Motor Skills, at post-intervention.

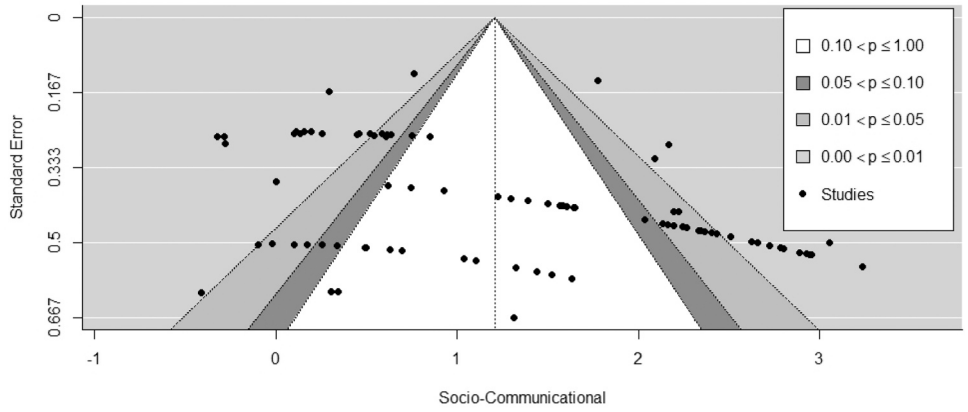


Fig. 7. Funnel Plot, Social and Communication Skills.

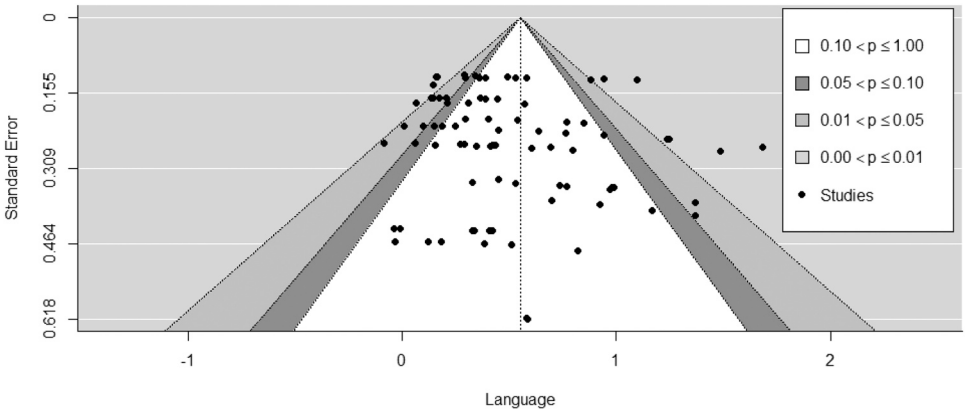


Fig. 8. Funnel Plot, Language Skills.

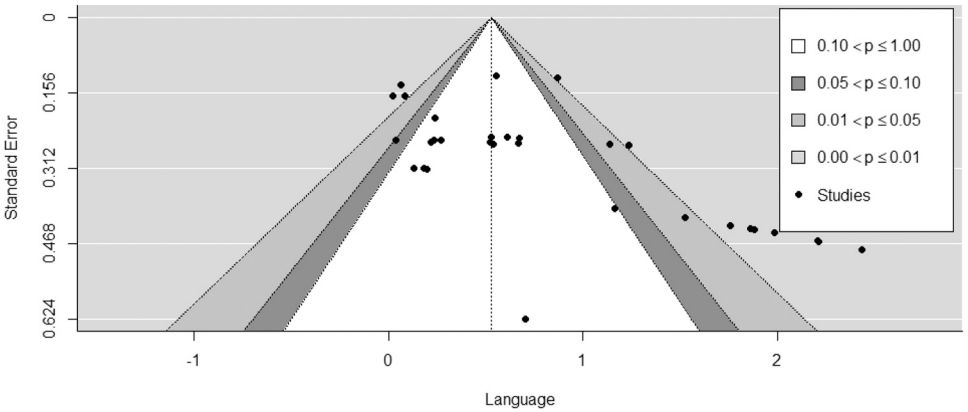


Fig. 9. Funnel Plot, Cognitive Skills.

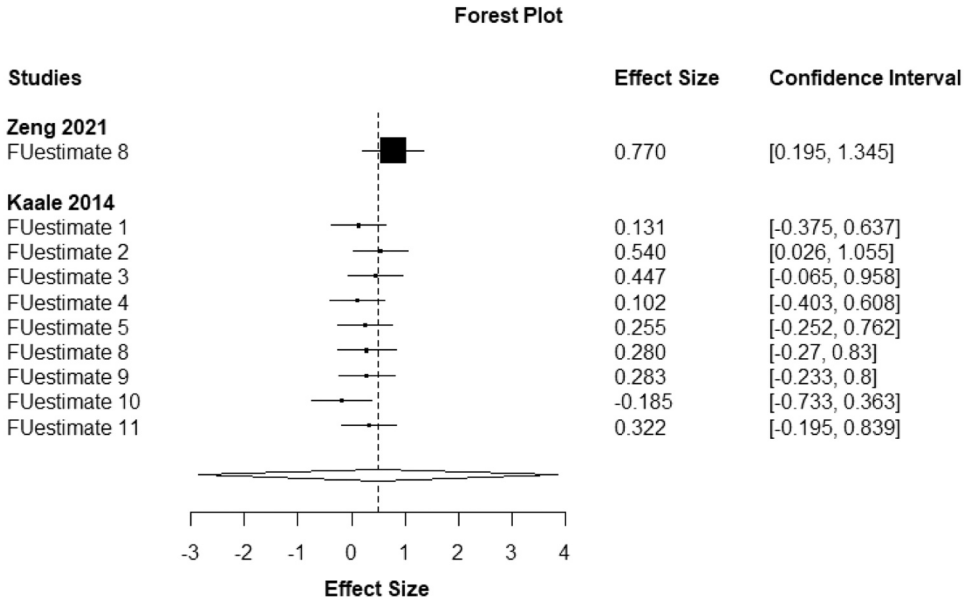


Fig. 10. Forest Plot, Social and Communicational Skills, at follow-up.

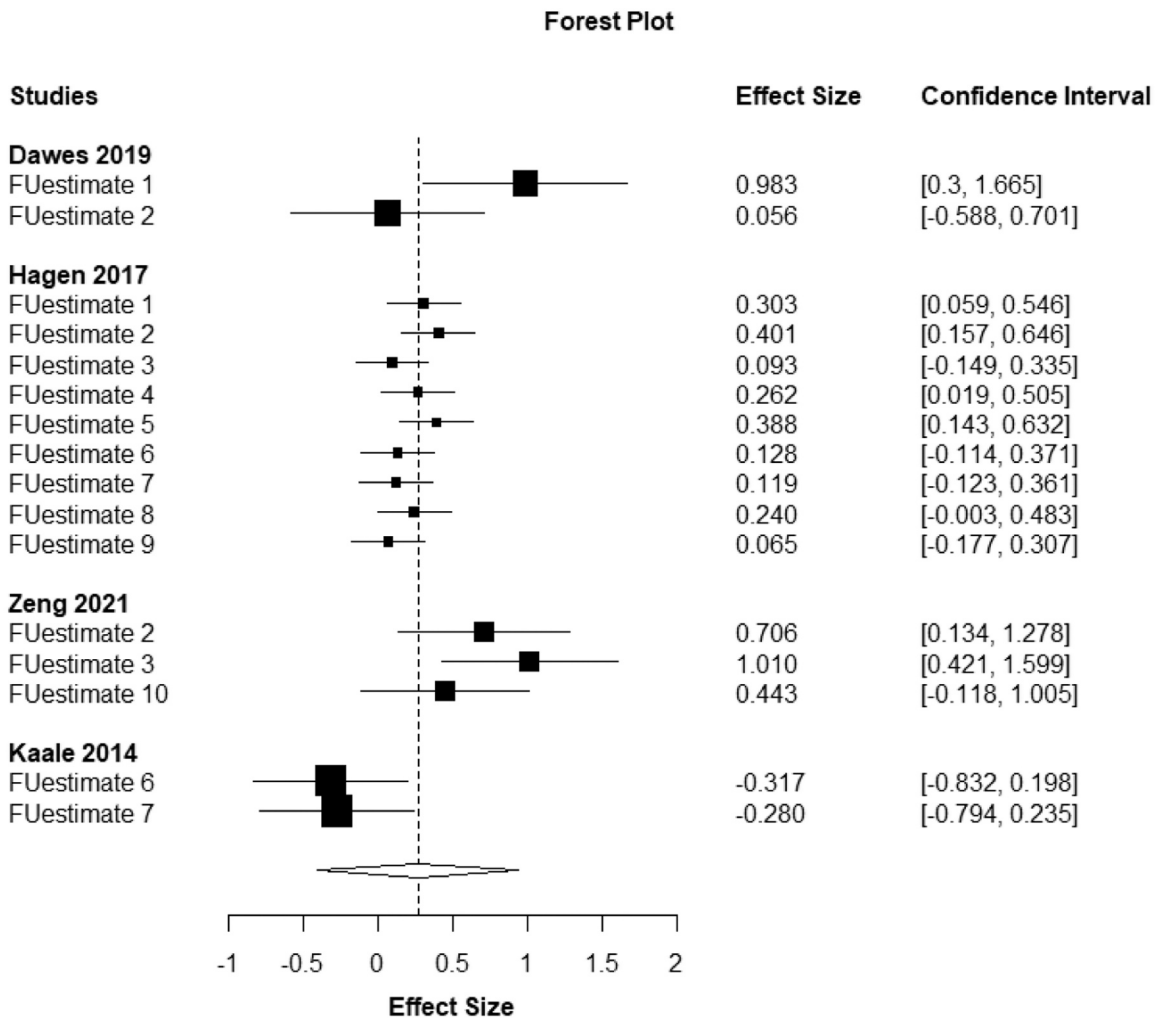


Fig. 11. Forest Plot, Language Skills, at follow-up.

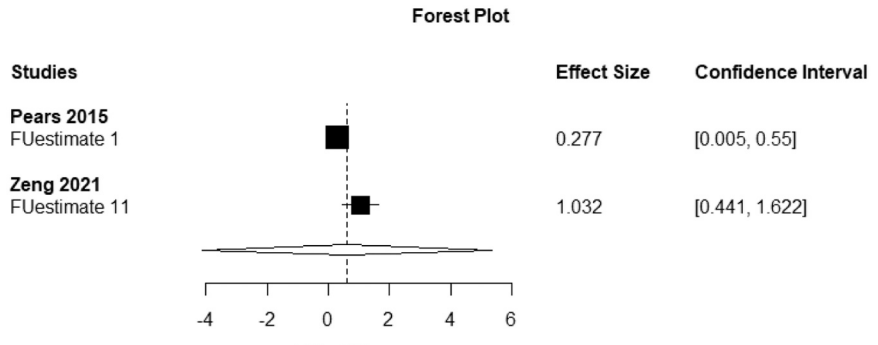


Fig. 12. Forest Plot, Emotional and Behavioral Adjustment, at follow-up.

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